

Trainings ▾

Select Service Tools

Recommended Cummins® Service Tools

- Diagnostic fuel line, Part Number 5299130.

Additional Service Items

- Container suitable for fuel
- 22 mm wrench.

General Information

The fuel system **must** be primed before starting the engine. If the fuel lines to the high-pressure fuel pump(s) or the pump(s) were removed, the pump(s) **must** also be primed.

Note : Controlled venting is provided at the injection pump through the fuel drain manifold. Small amounts of air introduced by changing the fuel filters or fuel injection pump supply line will be vented automatically if the fuel filter is changed in accordance with the instructions.

This procedure covers the fuel system with electric lift pumps. Refer to Procedure 006-024 in Section 6 (</qs3/pubsys2/xml/en/procedures/57/57-006-024-tr.html>) for mechanical lift pumps. See Vent step in procedure.

Initial Check

Verify the following:

- All customer and original equipment manufacturer (OEM) plumbing is connected
- OEM fuel tank is properly filled
- OEM fuel supply is open
- Batteries and power supplies are connected.

See equipment manufacturer service information.

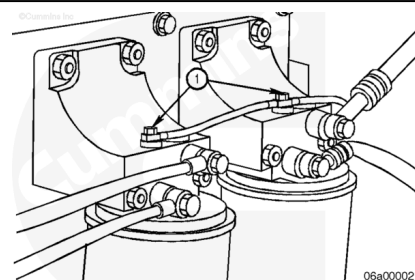
Prime

WARNING

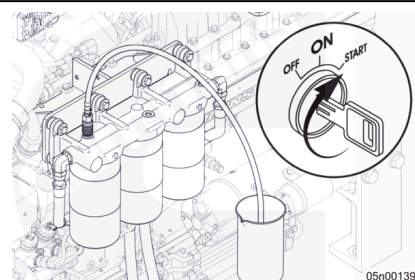
Depending on the circumstance, diesel fuel is flammable. When inspecting or performing service or repairs on the fuel system, to reduce the possibility of fire and resulting severe personal injury, death, or property damage, never smoke or allow sparks or flames (such as pilot lights, electrical switches, or welding equipment) in the work area.

Fuel System

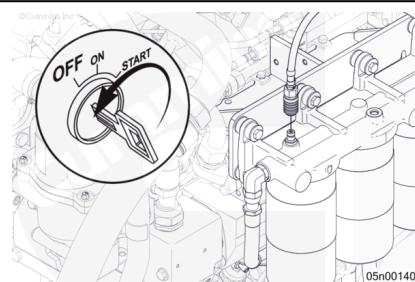
- Connect diagnostic fuel line, Part Number 5299130, or equivalent, to the Compuchek™ fitting on the stage 2 fuel filter head (1).
- Route the hose into a container suitable for fuel.



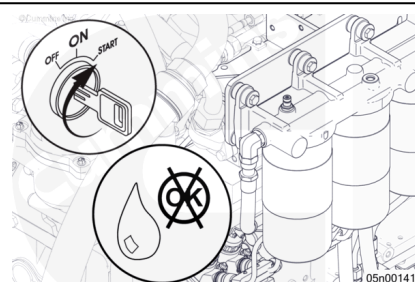
- Turn the keyswitch ON.
- Allow the fuel lift pump to run until fuel flows from the hose.
- Continue to cycle the keyswitch, if required.



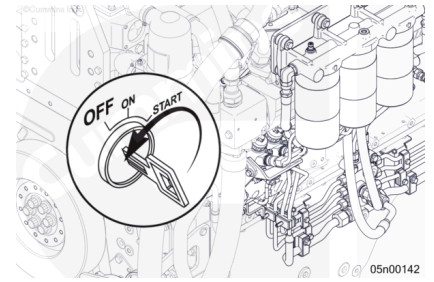
- Turn keyswitch OFF.
- Remove the diagnostic hose from the Compuchek™ fitting.



- Turn keyswitch ON.
- Allow the fuel lift pump to run until it shuts off. This can take up to 5 minutes.
- Check fuel system for leaks.



- Turn keyswitch OFF.



Fuel Pump

- **Note :** Each fuel pump must be primed prior to engine start. Prime the fuel pump farthest from the stage 2 fuel filter first.
- Loosen the banjo fitting (1) on the fuel injection supply line from the lift pump to the fuel pump. Use a 22 mm wrench. Do **not** remove the plug.
- Turn keyswitch ON to activate the lift pump.
- Allow the air to escape from the fitting. When a steady stream of fuel appears, close the banjo fitting.
- Turn keyswitch OFF.
- Tighten the banjo fitting to 10 N·m [120 in-lb].
- Operate the engine to 70°C [158°F] coolant temperature. Check for leaks.

